

Mathematical Analysis II

Week 9 Homework (April 28)

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Section 10.3

1. 计算下列三重积分.

(2) $\iiint_V xy^2z^3 \, dx dy dz$, V : 由 $z = xy$, $y = x$, $x = 1$, $z = 0$ 围成;

解. 该区域位于第一卦限, 积分区域可表示为 $0 \leq x \leq 1$, $0 \leq y \leq x$, $0 \leq z \leq xy$.
因此

$$\begin{aligned}\iiint_V xy^2z^3 \, dx dy dz &= \int_0^1 dx \int_0^x dy \int_0^{xy} xy^2z^3 \, dz \\ &= \frac{1}{4} \int_0^1 dx \int_0^x x^5 y^6 \, dy \\ &= \frac{1}{28} \int_0^1 x^{12} \, dx \\ &= \frac{1}{364}\end{aligned}$$

(4) $\iiint_V (a - y) \, dx dy dz$, V : 由 $y = 0$, $z = 0$, $2x + y = a$, $x + y = a$, $y + z = a$ 围成.

解. 积分区域满足 $0 \leq y \leq a$, $\frac{a-y}{2} \leq x \leq a-y$, $0 \leq z \leq a-y$.
因此

$$\begin{aligned}\iiint_V (a - y) \, dx dy dz &= \int_0^a dy \int_{\frac{a-y}{2}}^{a-y} dx \int_0^{a-y} (a - y) \, dz \\ &= \int_0^a dy \int_{\frac{a-y}{2}}^{a-y} (a - y)^2 \, dx \\ &= \frac{1}{2} \int_0^a (a - y)^3 \, dy \\ &= \frac{a^4}{8}\end{aligned}$$

2. 计算下列积分值.

$$(2) \int_{-R}^R dx \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} dy \int_0^{\sqrt{R^2-x^2-y^2}} (x^2 + y^2) dz;$$

解. 采用柱面坐标变换: 记 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$, 则积分区域为 $\begin{cases} 0 \leq r \leq R \\ 0 \leq \theta \leq 2\pi \\ 0 \leq z \leq \sqrt{R^2 - r^2} \end{cases}$, 雅可比行列

式为 $\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r$.

因此

$$\begin{aligned} \int_{-R}^R dx \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} dy \int_0^{\sqrt{R^2-x^2-y^2}} (x^2 + y^2) dz &= \int_0^R dr \int_0^{2\pi} d\theta \int_0^{\sqrt{R^2-r^2}} r^3 dz \\ &= \int_0^R dr \int_0^{2\pi} r^3 \sqrt{R^2 - r^2} d\theta \\ &= 2\pi \int_0^R r^3 \sqrt{R^2 - r^2} dr \end{aligned}$$

对 $\int r^3 \sqrt{R^2 - r^2} dr$ 采用分部积分法. 记 $u = r^2$, $dv = r \sqrt{R^2 - r^2} dr$, 则 $du = 2r dr$,
 $v = -\frac{1}{3}(R^2 - r^2)^{\frac{3}{2}}$.
 因此

$$\begin{aligned} \int r^3 \sqrt{R^2 - r^2} dr &= \int u dv \\ &= uv - \int v du \\ &= -\frac{r^2}{3}(R^2 - r^2)^{\frac{3}{2}} + \frac{2}{3} \int r(R^2 - r^2)^{\frac{3}{2}} dr \\ &= -\frac{r^2}{3}(R^2 - r^2)^{\frac{3}{2}} - \frac{2}{15}(R^2 - r^2)^{\frac{5}{2}} + C \end{aligned}$$

从而

$$\begin{aligned} \int_0^R r^3 \sqrt{R^2 - r^2} dr &= \left(-\frac{r^2}{3}(R^2 - r^2)^{\frac{3}{2}} - \frac{2}{15}(R^2 - r^2)^{\frac{5}{2}} \right) \Big|_0^R \\ &= \frac{2}{15} R^5 \end{aligned}$$

因此

$$\begin{aligned} \text{原式} &= 2\pi \int_0^R r^3 \sqrt{R^2 - r^2} dr \\ &= \frac{4}{15} \pi R^5 \end{aligned}$$

$$(4) \int_0^1 dx \int_0^{\sqrt{1-x^2}} dy \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} z^2 dz.$$

解. 该区域位于第一卦限.

采用柱面坐标变换: 记 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$, 则积分区域为 $\begin{cases} 0 \leq r \leq 1 \\ 0 \leq \theta \leq \frac{\pi}{2} \\ r \leq z \leq \sqrt{2-r^2} \end{cases}$, 雅可比行列

式为 $\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r$.
因此

$$\begin{aligned} \int_0^1 dx \int_0^{\sqrt{1-x^2}} dy \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} z^2 dz &= \int_0^1 dr \int_0^{\frac{\pi}{2}} d\theta \int_r^{\sqrt{2-r^2}} r z^2 dz \\ &= \frac{1}{3} \int_0^1 dr \int_0^{\frac{\pi}{2}} r \left[(2-r^2)^{\frac{3}{2}} - r^3 \right] d\theta \\ &= \frac{\pi}{6} \int_0^1 \left[r(2-r^2)^{\frac{3}{2}} - r^4 \right] dr \\ &= \frac{\pi}{6} \left[-\frac{1}{5}(2-r^2)^{\frac{5}{2}} - \frac{r^5}{5} \right]_0^1 \\ &= \frac{\pi}{15} (2\sqrt{2} - 1) \end{aligned}$$

3. 计算下列三重积分.

(2) $\iiint_V \sqrt{x^2 + y^2} dx dy dz$, V : 由 $x^2 + y^2 = z^2$, $z = 1$ 围成;

解. 该区域满足 $z \geq 0$.

采用柱面坐标变换: 记 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$, 则积分区域为 $\begin{cases} 0 \leq z \leq 1 \\ 0 \leq \theta \leq 2\pi \\ 0 \leq r \leq z \end{cases}$, 雅可比行列式为

$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r$.
因此

$$\begin{aligned} \iiint_V \sqrt{x^2 + y^2} dx dy dz &= \int_0^1 dz \int_0^{2\pi} d\theta \int_0^z r^2 dr \\ &= \frac{1}{3} \int_0^1 dz \int_0^{2\pi} z^3 d\theta \\ &= \frac{2\pi}{3} \int_0^1 z^3 dz \\ &= \frac{\pi}{6} \end{aligned}$$

(4) $\iiint_V xyz \, dx dy dz$, V : 是 $x^2 + y^2 + z^2 \leq 1$ 的第一卦限部分;

解. 采用球面坐标变换: 记 $\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$, 则积分区域为 $\begin{cases} 0 \leq r \leq 1 \\ 0 \leq \theta \leq \frac{\pi}{2} \\ 0 \leq \varphi \leq \frac{\pi}{2} \end{cases}$, 雅可比行列式

为 $\frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} = r^2 \sin \theta$.
因此

$$\begin{aligned} \iiint_V xyz \, dx dy dz &= \int_0^1 r^5 \, dr \int_0^{\frac{\pi}{2}} \sin^3 \theta \cos \theta \, d\theta \int_0^{\frac{\pi}{2}} \sin \varphi \cos \varphi \, d\varphi \\ &= \frac{1}{48} \end{aligned}$$

(6) $\iiint_V |x^2 + y^2 + z^2 - 1| \, dx dy dz$, V : $x^2 + y^2 + z^2 \leq 4$;

解. 由于被积函数关于原点中心对称, 可先计算第一卦限的积分再乘以 8.

采用球面坐标变换: 记 $\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$, 则积分区域为 $\begin{cases} 0 \leq r \leq 2 \\ 0 \leq \theta \leq \frac{\pi}{2} \\ 0 \leq \varphi \leq \frac{\pi}{2} \end{cases}$, 雅可比行列式

为 $\frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} = r^2 \sin \theta$.
因此

$$\begin{aligned} \iiint_V |x^2 + y^2 + z^2 - 1| \, dx dy dz &= 8 \int_0^2 dr \int_0^{\frac{\pi}{2}} d\theta \int_0^{\frac{\pi}{2}} |r^2 - 1| r^2 \sin \theta \, d\varphi \\ &= 8 \left(\int_0^1 (r^2 - r^4) \, dr + \int_1^2 (r^4 - r^2) \, dr \right) \int_0^{\frac{\pi}{2}} \sin \theta \, d\theta \int_0^{\frac{\pi}{2}} d\varphi \\ &= 16\pi \end{aligned}$$

(8) $\iiint_V (|x| + z) e^{-(x^2 + y^2 + z^2)} \, dx dy dz$, V : $1 \leq x^2 + y^2 + z^2 \leq 4$.

解. 由于区域关于平面 $x = 0$ 对称, 且被积函数中 $|x|$ 为偶函数, 可先计算 $x \geq 0, y \geq 0$ 部分的积分再乘以 4.

采用球面坐标变换: 记 $\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$, 则积分区域为 $\begin{cases} 1 \leq r \leq 2 \\ 0 \leq \theta \leq \pi \\ 0 \leq \varphi \leq \frac{\pi}{2} \end{cases}$, 雅可比行列式

为 $\frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} = r^2 \sin \theta$.
因此

$$\begin{aligned}
\iiint_V (|x| + z) e^{-(x^2+y^2+z^2)} dx dy dz &= 4 \int_1^2 r^3 e^{-r^2} dr \int_0^\pi d\theta \int_0^{\frac{\pi}{2}} (\sin \theta \cos \varphi + \cos \theta) \sin \theta d\varphi \\
&= 4 \int_1^2 r^3 e^{-r^2} dr \int_0^\pi \left(\sin^2 \theta \cdot \frac{\pi}{2} + \frac{\pi}{2} \sin \theta \cos \theta \right) d\theta \\
&= 2\pi \int_1^2 r^3 e^{-r^2} dr \int_0^\pi \sin^2 \theta d\theta \\
&= 2\pi \cdot \frac{2e^3 - 5}{2e^4} \cdot \frac{\pi}{2} \\
&= \frac{\pi^2(2e^3 - 5)}{2e^4}
\end{aligned}$$

4. 利用对称性求下列三重积分.

(2) $\iiint_V x dx dy dz$, V : 由 $x^2 + y^2 = z^2$, $x^2 + y^2 = 1$ 围成;

解. 采用柱面坐标变换: 记 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$, 则积分区域为 $\begin{cases} 0 \leq r \leq 1 \\ 0 \leq \theta \leq 2\pi \\ -r \leq z \leq r \end{cases}$, 雅可比行列式为

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r.$$

由于该区域为对称双锥, 且被积函数 x 关于 x 为奇函数, 因此

$$\begin{aligned}
\iiint_V x dx dy dz &= \int_0^1 r^2 dr \int_0^{2\pi} \cos \theta d\theta \int_{-r}^r dz \\
&= 0
\end{aligned}$$

(4) $\iiint_V (x^2 + y^2) dx dy dz$, V : $r^2 \leq x^2 + y^2 + z^2 \leq R^2$, $z \geq 0$;

解. 采用球面坐标变换: 记 $\begin{cases} x = \rho \sin \theta \cos \varphi \\ y = \rho \sin \theta \sin \varphi \\ z = \rho \cos \theta \end{cases}$, 则积分区域为 $\begin{cases} r \leq \rho \leq R \\ 0 \leq \theta \leq \frac{\pi}{2} \\ 0 \leq \varphi \leq 2\pi \end{cases}$, 雅可比行列式

$$\text{为 } \frac{\partial(x, y, z)}{\partial(\rho, \theta, \varphi)} = \rho^2 \sin \theta.$$

因此

$$\begin{aligned}
\iiint_V (x^2 + y^2) dx dy dz &= \int_r^R \rho^4 d\rho \int_0^{\frac{\pi}{2}} \sin^3 \theta d\theta \int_0^{2\pi} d\varphi \\
&= \frac{2\pi}{5} \cdot (R^5 - r^5) \cdot \left(-\cos \theta + \frac{1}{3} \cos^3 \theta \right) \Big|_0^{\frac{\pi}{2}} \\
&= \frac{4\pi}{15} (R^5 - r^5)
\end{aligned}$$

$$(6) \iiint_V \frac{z \ln(x^2 + y^2 + z^2 + 1)}{x^2 + y^2 + z^2 + 1} dx dy dz, V: x^2 + y^2 + z^2 \leq 1;$$

解. 注意到被积函数关于 z 为奇函数, 而积分区域关于平面 $z = 0$ 对称, 因此原积分值为 0.

$$\text{验证: 采用球面坐标变换, 记 } \begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}, \text{ 则积分区域为 } \begin{cases} 0 \leq r \leq 1 \\ 0 \leq \theta \leq \pi \\ 0 \leq \varphi \leq 2\pi \end{cases}, \text{ 雅可比}$$

$$\text{行列式为 } \frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} = r^2 \sin \theta.$$

因此

$$\iiint_V \frac{z \ln(x^2 + y^2 + z^2 + 1)}{x^2 + y^2 + z^2 + 1} dx dy dz = \int_0^1 \frac{r^3 \ln(r^2 + 1)}{r^2 + 1} dr \int_0^\pi \cos \theta \sin \theta d\theta \int_0^{2\pi} d\varphi,$$

$$\text{其中 } \int_0^\pi \cos \theta \sin \theta d\theta = 0.$$

$$\text{因此 } \iiint_V \frac{z \ln(x^2 + y^2 + z^2 + 1)}{x^2 + y^2 + z^2 + 1} dx dy dz = 0.$$

(8) 书上没这道题目.

5. 计算下列曲面围成的立体体积.

$$(1) y = 0, z = 0, 3x + y = 6, 3x + 2y = 12, x + y + z = 6;$$

$$\text{解. 积分区域可表示为 } \begin{cases} 0 \leq y \leq 6 \\ 2 - \frac{y}{3} \leq x \leq 4 - \frac{2y}{3} \\ 0 \leq z \leq 6 - x - y \end{cases}.$$

因此体积为

$$\begin{aligned} \int_0^6 dy \int_{2-\frac{y}{3}}^{4-\frac{2y}{3}} dx \int_0^{6-x-y} dz &= \int_0^6 dy \int_{2-\frac{y}{3}}^{4-\frac{2y}{3}} (6-x-y) dx \\ &= \int_0^6 \left(6x - \frac{x^2}{2} - yx \right) \Big|_{2-\frac{y}{3}}^{4-\frac{2y}{3}} dy \\ &= \int_0^6 \left(\frac{y^2}{6} - 2y + 6 \right) dy \\ &= \left(\frac{y^3}{18} - y^2 + 6y \right) \Big|_0^6 \\ &= 12 \end{aligned}$$

$$(3) z^2 + x^2 = 1 \text{ 和 } x + y + z = 3, y = 0;$$

$$\text{解. 采用柱面坐标变换: 记 } \begin{cases} x = r \cos \theta \\ z = r \sin \theta \\ y = y \end{cases}, \text{ 则积分区域为 } \begin{cases} 0 \leq r \leq 1 \\ 0 \leq \theta \leq 2\pi \\ 0 \leq y \leq 3 - r(\cos \theta + \sin \theta) \end{cases}, \text{ 雅可比}$$

$$\text{可比行列式为 } \frac{\partial(x, y, z)}{\partial(r, \theta, y)} = r.$$

因此体积为

$$\begin{aligned}\int_0^1 dr \int_0^{2\pi} (3 - r(\cos \theta + \sin \theta))r d\theta &= \int_0^1 (3\theta - r \sin \theta + r \cos \theta) r \Big|_0^{2\pi} dr \\ &= \int_0^1 6\pi r dr \\ &= 3\pi\end{aligned}$$

(5) $\frac{x^2}{9} + \frac{y^2}{4} = 1, z = xy$ (在第一卦限部分);

解. 采用广义柱面坐标变换: 记 $\begin{cases} x = 3r \cos \theta \\ y = 2r \sin \theta \\ z = z \end{cases}$, 则积分区域为 $\begin{cases} 0 \leq r \leq 1 \\ 0 \leq \theta \leq \frac{\pi}{2} \\ 0 \leq z \leq 6r^2 \cos \theta \sin \theta \end{cases}$, 雅

可比行列式为 $\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = 6r$.

因此体积为

$$36 \int_0^1 r^3 dr \int_0^{\frac{\pi}{2}} \cos \theta \sin \theta d\theta = \frac{9}{2}$$

(7) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$ (含 z 轴部分).

解. 采用广义球面坐标变换: 记 $\begin{cases} x = ar \sin \theta \cos \varphi \\ y = br \sin \theta \sin \varphi \\ z = cr \cos \theta \end{cases}$, 则积分区域为 $\begin{cases} 0 \leq r \leq 1 \\ 0 \leq \theta \leq \pi \\ 0 \leq \varphi \leq 2\pi \end{cases}$, 雅可比

行列式为 $\frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} = abcr^2 \sin \theta$.

由于该区域包含 z 轴, 需满足 $\sin^2 \theta \leq \cos^2 \theta$, 解得 $0 \leq \theta \leq \frac{\pi}{4}$ 或 $\frac{3\pi}{4} \leq \theta \leq \pi$.

由于两个区域体积相等, 可先计算一个区域的体积再乘以 2.

因此体积为

$$\begin{aligned}2abc \cdot 2\pi \int_0^1 r^2 dr \int_0^{\frac{\pi}{4}} \sin \theta d\theta &= \frac{4}{3}abc\pi \left(1 - \frac{\sqrt{2}}{2}\right) \\ &= \frac{4 - 2\sqrt{2}}{3}abc\pi\end{aligned}$$

7. 设 $F(t) = \iiint_{x^2+y^2+z^2 \leq t^2} f(x^2 + y^2 + z^2) dx dy dz$, 其中 f 为可微分函数, 求 $F'(t)$.

解. 采用球面坐标变换: 记 $\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$, 则积分区域为 $\begin{cases} 0 \leq r \leq t \\ 0 \leq \theta \leq \pi \\ 0 \leq \varphi \leq 2\pi \end{cases}$, 雅可比行列式

为 $\frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} = r^2 \sin \theta$.
因此

$$\begin{aligned} \iiint_{x^2+y^2+z^2 \leq t^2} f(x^2+y^2+z^2) dx dy dz &= \int_0^t f(r^2) r^2 dr \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\varphi \\ &= 4\pi \int_0^t f(r^2) r^2 dr \end{aligned}$$

由变上限积分求导法则可得 $F'(t) = 4\pi t^2 f(t^2)$.

8. 证明: $\iiint_{x^2+y^2+z^2 \leq 1} f(z) dv = \pi \int_{-1}^1 f(z)(1-z^2) dz.$

证. 考虑等式左边的三重积分, 先对 x, y 积分.

对于固定的 z , 截面区域为 $x^2 + y^2 \leq 1 - z^2$. 采用柱面坐标变换: 记 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$, 则 $\begin{cases} 0 \leq r \leq \sqrt{1-z^2} \\ 0 \leq \theta \leq 2\pi \end{cases}$, 雅可比行列式为 $\frac{\partial(x, y)}{\partial(r, \theta)} = r$.
因此

$$\begin{aligned} \iiint_{x^2+y^2+z^2 \leq 1} f(z) dx dy dz &= \int_{-1}^1 f(z) dz \iint_{x^2+y^2 \leq 1-z^2} dx dy \\ &= \int_{-1}^1 f(z) dz \int_0^{\sqrt{1-z^2}} r dr \int_0^{2\pi} d\theta \\ &= \pi \int_{-1}^1 f(z)(1-z^2) dz \end{aligned}$$

这与等式右边相等.

Q.E.D.

12. 一个物体是由两个半径各为 R 和 r ($R > r$) 的同心球所围成, 已知材料的密度和到球心的距离成反比, 且在距离等于 1 处的密度等于 k , 求物体的总质量.

解. 采用球面坐标变换: 记 $\begin{cases} x = \rho \sin \theta \cos \varphi \\ y = \rho \sin \theta \sin \varphi \\ z = \rho \cos \theta \end{cases}$, 则体积元为 $\rho^2 \sin \theta d\rho d\theta d\varphi$.

由题意可知, 在点 (ρ, θ, φ) 处的密度为 $\frac{k}{\rho}$.

因此物体的总质量为

$$2\pi \int_r^R \frac{k}{\rho} \cdot \rho^2 d\rho \int_0^\pi \sin \theta d\theta = 2k\pi(R^2 - r^2)$$

13. 半径为 a 的圆盘, 其各点的密度和到圆心的距离成正比 (比例系数为 1), 今内切于圆盘截去半径为 $\frac{a}{2}$ 之小圆, 求余下部分的重心坐标.

解. 由对称性可知, 重心位于两圆圆心连成的直线上. 以该直线为 x 轴, 大圆圆心为坐标原点, 垂直方向为 y 轴建立坐标系, 则只需计算重心在 x 轴方向上的坐标.

坐标 (x, y) 处的密度为 $\sqrt{x^2 + y^2}$.

需要计算剩余部分的力矩

$$\iint_{x^2+y^2 \leq a^2} x\sqrt{x^2+y^2} dx dy - \iint_{(x-\frac{a}{2})^2+y^2 \leq \frac{a^2}{4}} x\sqrt{x^2+y^2} dx dy$$

和质量

$$\iint_{x^2+y^2 \leq a^2} \sqrt{x^2+y^2} dx dy - \iint_{(x-\frac{a}{2})^2+y^2 \leq \frac{a^2}{4}} \sqrt{x^2+y^2} dx dy$$

采用柱面坐标变换: 记 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$, 则雅可比行列式为 $\frac{\partial(x, y)}{\partial(r, \theta)} = r$.

对于大圆区域, 积分区域为 $\begin{cases} 0 \leq r \leq a \\ 0 \leq \theta \leq 2\pi \end{cases}$.

对于小圆区域, 由 $(x - \frac{a}{2})^2 + y^2 \leq \frac{a^2}{4}$ 可得 $r^2 - ar \cos \theta + \frac{a^2}{4} \leq \frac{a^2}{4}$, 即 $r \leq a \cos \theta$. 因此

积分区域为 $\begin{cases} -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \\ 0 \leq r \leq a \cos \theta \end{cases}$.

首先计算大圆的力矩和质量:

$$\begin{aligned} \text{力矩} &= \iint_{x^2+y^2 \leq a^2} x\sqrt{x^2+y^2} dx dy \\ &= \int_0^a r^3 dr \int_0^{2\pi} \cos \theta d\theta \\ &= \frac{a^4}{4} \cdot 0 \\ &= 0 \end{aligned}$$

这是符合直觉的.

$$\begin{aligned}
\text{质量} &= \iint_{x^2+y^2 \leq a^2} \sqrt{x^2+y^2} \, dx dy \\
&= \int_0^a r^2 \, dr \int_0^{2\pi} d\theta \\
&= \frac{2a^3\pi}{3}
\end{aligned}$$

接下来计算小圆的力矩:

$$\begin{aligned}
\text{力矩} &= \iint_{\left(x-\frac{a}{2}\right)^2+y^2 \leq \frac{a^2}{4}} x\sqrt{x^2+y^2} \, dx dy \\
&= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{a \cos \theta} r^3 \cos \theta \, dr \\
&= \frac{a^4}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^5 \theta \, d\theta
\end{aligned}$$

计算 $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^5 \theta \, d\theta$: 由于被积函数为偶函数, 因此

$$\begin{aligned}
\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^5 \theta \, d\theta &= 2 \int_0^{\frac{\pi}{2}} \cos^5 \theta \, d\theta \\
&= 2 \int_0^1 (1-u^2)^2 \, du \\
&= 2 \int_0^1 (1-2u^2+u^4) \, du \\
&= 2 \left(u - \frac{2}{3}u^3 + \frac{1}{5}u^5 \right) \Big|_0^1 \\
&= \frac{16}{15}
\end{aligned}$$

其中采用了换元法: 记 $u = \sin \theta$, 则 $du = \cos \theta \, d\theta$.

因此小圆的力矩为 $\frac{4a^4}{15}$.

再计算小圆的质量:

$$\begin{aligned}
\text{质量} &= \iint_{\left(x-\frac{a}{2}\right)^2+y^2\leq\frac{a^2}{4}} \sqrt{x^2+y^2} \, dx dy \\
&= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{a\cos\theta} r^2 \, dr \\
&= \frac{a^3}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3\theta \, d\theta \\
&= \frac{4a^3}{9}
\end{aligned}$$

因此剩余部分的重心 x 坐标为

$$\frac{0 - \frac{4a^4}{15}}{\frac{2a^3\pi}{3} - \frac{4a^3}{9}} = \frac{6a}{10 - 15\pi}$$

即重心坐标为 $\left(\frac{6a}{10 - 15\pi}, 0\right)$.

第 10 章综合习题

2. 计算三重积分

$$I = \iiint_{[0,1]^3} \frac{du dv dw}{(1 + u^2 + v^2 + w^2)^2}.$$

解.

3. 设 $a > 0, b > 0$. 试求下面的积分:

$$(1) I_1 = \int_0^1 \sin\left(\ln \frac{1}{x}\right) \cdot \frac{x^b - x^a}{\ln x} \, dx;$$

解.

$$(2) I_2 = \int_0^1 \cos\left(\ln \frac{1}{x}\right) \cdot \frac{x^b - x^a}{\ln x} \, dx.$$

解.

$$4. \text{ 设 } D = \{(x, y) \mid x^2 + y^2 \leq 1\}. \text{ 求 } I = \iint_D \left| \frac{x+y}{\sqrt{2}} - x^2 - y^2 \right| \, dx dy.$$

解.